

Launch Angle, Speed and Range: Analysing Projectile Motion

PhET Projectile Motion · PhET

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/projectile-motion>

Curriculum links: forces · motion

Learning intentions

- I can investigate how launch angle affects the range of a projectile at constant speed.
- I can use projectile motion equations to calculate range, maximum height and time of flight.
- I can explain the effect of air resistance on a projectile's trajectory.

Getting started

Open PhET Projectile Motion and use the cannon on the Intro or Lab screen. Set a fixed launch speed and keep air resistance turned off for the first tasks — you will turn it on later to compare.

Tasks

1. Before testing, predict which launch angle will give the maximum range for a constant launch speed. Write down your prediction.

2. Test launch angles of roughly 15, 30, 45, 60 and 75 degrees at the same speed, and record the range, maximum height and time of flight for each.

Launch angle	Range	Max height	Time of flight

3. Using your table, describe the pattern you found between launch angle and range. Was your prediction correct?

4. For a launch speed of 20 m/s at 45 degrees, calculate the expected time of flight using kinematics equations, ignoring air resistance. Compare your calculated value with the simulation's reading.

Working:

5. For the same launch, 20 m/s at 45 degrees, calculate the theoretical maximum range using the standard range equation, and compare it with the simulation.

Working:

6. Turn on air resistance and repeat the 45-degree launch at the same speed. How do the range and time of flight change compared with your earlier calculation?

7. Explain why real-world projectiles, such as a thrown javelin or a kicked ball, don't perfectly match the 'no air resistance' equations, using evidence from your air-resistance comparison.

Extension (optional)

A cannonball and a beach ball are launched at the same angle and speed, one with air resistance on and one off. Predict and explain which will land further away, and why the effect of air resistance differs so much between the two objects.
